THE COMPARISON FUNCTOR IS AN EQUIVALENCE FOR QUASI-COMPACT SCHEMES WITH AFFINE DIAGONAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A morphism from an affine scheme to a scheme with affine diagonal is affine. invoked at proof lines 1813-1815. Makes each affine open U -> X an affine morphism.
- The affine pushforward lemma verifies the coherator hypothesis of the abstract argument lemma. invoked at proof lines 1816-1817. Concludes the equivalence.

Source reference. Original equation 06VT: sources/stacks-perfect/pinned-original-source.tex, source lines 289–289. Cited at statement line 1803 as the functor shown to be an equivalence; the proof is inside the counted span.

Source reference. Original lemma 01SG: sources/stacks-chow/tags.txt, source lines 2337–2337. Cited at proof lines 1814-1815 for the affineness of U -> X; the application is inside the counted span.

Source reference. Original lemma 09T6: sources/stacks-perfect/pinned-original-source.tex, source lines 1702–1702. Cited at proof lines 1816-1816 as the final input; the hypothesis verification is inside the counted span.

Source reference. Original lemma 08D7: sources/stacks-perfect/pinned-original-source.tex, source lines 1580–1580. Cited at proof lines 1817-1817 to verify the hypothesis; the conclusion is inside the counted span.

Proposition 0.2. *Let X be a quasi-compact scheme with affine diagonal. Then the functor (06VT)*

$$D(QCoh(\mathcal{O}_X)) \longrightarrow D_{QCoh}(\mathcal{O}_X)$$

is an equivalence with quasi-inverse given by RQ_X.

Proof. Let $U \subset X$ be an affine open. Then the morphism $U \to X$ is affine by Morphisms, Lemma 01SG. Thus the assumption of Lemma 09T6 holds by Lemma 08D7 and we win. □

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